Counterexamples to Erdős Problem 1075

Qiyuan Gu
University of Chicago
phoenix1203@uchicago.edu

Abstract

Erdős conjectured that, for every fixed $r \geq 3$, there is a constant $c_r > r^{-r}$ such that, for every $\varepsilon > 0$, every sufficiently large r-uniform hypergraph on N vertices with at least $(1+\varepsilon)(N/r)^r$ edges contains a subgraph on $m \to \infty$ vertices with at least $c_r m^r$ edges. We give counterexamples for every $r \geq 5$. The base construction is a sequence of explicit 5-uniform hypergraphs G_n whose unnormalized Lagrangians satisfy

$$5^{-5}\left(1 + \frac{1}{4(3n)^3}\right) \leq \lambda(G_n) \leq 5^{-5} + \frac{1}{4!n}.$$

Blow-ups give the counterexample for $r = 5$, and adjoining common vertices to every edge lifts it to all larger uniformities. Thus Problem 1075, as a statement for all $r \geq 3$, has a negative answer. The endpoint question remains open for $3 \leq r \leq 4$, including the case 2/9 for 3-graphs. The proof is formalized and verified in Lean 4.

1 Introduction

All hypergraphs in this paper are finite and simple. For a hypergraph H, write $v(H) = |V(H)|$, $e(H) = |E(H)|$, and let $H[S]$ denote the subgraph induced by $S \subseteq V(H)$.

Erdős proved that, for every fixed $r \geq 2$ and $\varepsilon > 0$, an r-uniform hypergraph on N vertices with at least εN^r edges contains a complete r-partite r-uniform hypergraph with equal part sizes tending to infinity with N [1]. In particular, it contains a subgraph on $m \to \infty$ vertices with $(m/r)^r$ edges. Erdős subsequently conjectured that, for each fixed $r \geq 3$, there is a constant $c_r > r^{-r}$ such that, for every $\varepsilon > 0$ and all sufficiently large N, the assumption

$$e(H) \geq (1+\varepsilon)(N/r)^r$$

forces a subgraph on $m = m(N) \to \infty$ vertices with at least $c_r m^r$ edges [3, p. 80]. This is Problem 1075 in Bloom's collection [13].

Use the density $d(H) = e(H)/\binom{v(H)}{r}$. A number $\alpha \in [0,1)$ is a *jump* for r-graphs if there is a $c > 0$ such that, for every $\varepsilon > 0$ and integer $m \geq r$, all sufficiently large r-graphs of density at least $\alpha + \varepsilon$ contain an m-vertex subgraph of density at least $\alpha + c$; otherwise it is a *non-jump* [4]. The threshold $(N/r)^r$ corresponds asymptotically to $\alpha_r = r!/r^r$. Erdős proved that every number in $[0, \alpha_r)$ is a jump for r-graphs [2]. Thus Problem 1075 is the endpoint question at α_r. Frankl and Rödl disproved the earlier jumping constant conjecture by constructing non-jumps for every $r \geq 3$ [4, Theorem 1.2]. Frankl, Peng, Rödl and Talbot later showed that $(5/2)r!/r^r$ is a non-jump for every $r \geq 3$ [5], and Peng proved the lifting theorem that carries a non-jump of the form $c\,r!/r^r$ to all larger uniformities [6, Theorem 1]. Yan and Peng proved that 12/25 is a non-jump for 3-graphs, yielding $(54/25)r!/r^r$ for every $r \geq 3$ by lifting [7].

For $r \geq 4$, Shaw proved that $2r!/r^r$ is a non-jump and is the smallest non-jump obtainable from his finite-pattern formulation of the Frankl–Rödl method [8, Theorems 4.3 and 4.4(ii)]. For $r = 3$, the corresponding threshold is $6(5\sqrt{5} - 2)/121$ [8, Theorem 4.4(i)]. Liu and Mubayi subsequently proved that $4/9$ is a non-jump for 3-graphs, below this threshold [9]. The endpoint $2/9$ for 3-graphs remains open; Baber and Talbot proved, using flag algebras, jump intervals above it, the first beginning at 0.2299 [10]. The result below shows that the endpoint is a non-jump for every $r \geq 5$; the cases $3 \leq r \leq 4$ remain open.

Theorem 1. *For every $\gamma > 5^{-5}$, there is an $\varepsilon > 0$ and there are arbitrarily large 5-uniform hypergraphs H such that*

$$e(H) \geq (1+\varepsilon)\big(v(H)/5\big)^5, \qquad e(H[S]) < \gamma |S|^5 \quad \textit{for every nonempty } S \subseteq V(H).$$

For every nonempty vertex set S, any subgraph on S has at most $e(H[S])$ edges, so the theorem bounds all subgraphs on that vertex set. Taking $5^{-5} < \gamma < c_5$ in Theorem 1 rules out any $c_5 > 5^{-5}$. In the usual density normalization, it also implies that $5!/5^5$ is a non-jump. The proof uses explicit finite hypergraphs whose Lagrangians approach 5^{-5} from above. We use the unnormalized Lagrangian of [4, Section 2],

$$P_H(x) = \sum_{e \in E(H)} \prod_{v \in e} x_v, \qquad \lambda(H) = \max_{\substack{x_v \geq 0 \\ \sum_v x_v = 1}} P_H(x), \tag{1.1}$$

without the factor $r!$; see also [11, 12].

Shaw's criterion [8, Theorem 3.1] uses a finite r-pattern P, whose edges may repeat vertices, and a vertex v with positive weight in a maximizing weighting x, such that $\lambda(\mathrm{FR}_v(P)) = \lambda(P) < 1$. For a simple P, the construction $\mathrm{FR}_v(P)$ replaces v by r clones and adds the edge consisting of those clones. Splitting x_v equally among them preserves the weight of the old edges and gives

$$\lambda(\mathrm{FR}_v(P)) \geq \lambda(P) + (x_v/r)^r > \lambda(P).$$

Thus the equality condition fails for each of the simple hypergraphs G_n used here. Instead, we bound their full Lagrangians as n tends to infinity.

Proposition 2. *For every integer $n \geq 2$, the 5-uniform hypergraph G_n defined in Section 2 has $2n + 7$ vertices and satisfies*

$$5^{-5}\left(1 + \frac{1}{4(3n)^3}\right) \leq \lambda(G_n) \leq 5^{-5} + \frac{1}{4!n}. \tag{1.2}$$

The mechanism is visible at the level of the cubic forms. After normalizing the bottom mass, the balanced weighting on the closed cycle has

$$b_i + c = B - b_i + D = a_{i+1} + c = A - a_{i+1} + D = \frac{1}{3},$$

so both cubic forms attain $1/27$ even with $D > 0$, while the additional edge $\{U, V\} \cup \mathcal{D}$ contributes a gain of order D^3. Deleting one vertex A_k opens the cycle; the terminal mismatch forces a loss of order D^3 with a coefficient large enough to absorb that gain. Consequently every open path T_L has Lagrangian exactly 5^{-5}, whereas the closed cycle has Lagrangian slightly above 5^{-5}. Since a long cycle always has a vertex of small weight, the passage from a cycle to a path also gives an upper bound approaching the same threshold.

2 The cyclic construction

Fix $n \geq 2$. Take pairwise distinct vertices

$$A_i, B_i \quad (i \in \mathbb{Z}/n\mathbb{Z}), \qquad C, U, V, \qquad W, \qquad D_1, D_2, D_3.$$

Put $\mathcal{D} = \{D_1, D_2, D_3\}$. On the vertex set $\{A_i, B_i : i \in \mathbb{Z}/n\mathbb{Z}\} \cup \{C\} \cup \mathcal{D}$, define two three-uniform hypergraphs by

$$E(H_1) = \bigcup_i \{\{A_i, X, Y\} : X \in \{B_i, C\},\ Y \in \{B_j : j \neq i\} \cup \mathcal{D}\},$$
$$E(H_2) = \bigcup_i \{\{B_i, X, Y\} : X \in \{A_{i+1}, C\},\ Y \in \{A_j : j \neq i+1\} \cup \mathcal{D}\}.$$

All subscripts are read modulo n. Define

$$\begin{aligned} E(G_n) = \{&\{W, U\} \cup e : e \in E(H_1)\} \\ &\cup \{\{W, V\} \cup e : e \in E(H_2)\} \cup \{\{U, V\} \cup \mathcal{D}\}. \end{aligned} \tag{2.1}$$

Each triple in H_1 has a unique A-vertex, which determines i; the disjoint sets of choices for X and Y then determine both vertices uniquely. The same argument applies to H_2, using its unique B-vertex. Thus these parameterizations count each triple once. Every edge of G_n has five distinct vertices, and the three edge families are distinguished by their intersections with $\{U, V\}$. Hence G_n is a simple 5-uniform hypergraph, with no repeated terms in the polynomial below.

Use lower-case letters for the weights of the corresponding vertices, and write

$$A = \sum_i a_i, \qquad B = \sum_i b_i, \qquad D = \sum_{j=1}^{3} d_j.$$

The two cubic forms and the polynomial of G_n are

$$P_1 = \sum_i a_i(b_i + c)(B - b_i + D), \tag{2.2}$$

$$P_2 = \sum_i b_i(a_{i+1} + c)(A - a_{i+1} + D), \tag{2.3}$$

$$P_{G_n} = w(uP_1 + vP_2) + uv \prod_{j=1}^{3} d_j. \tag{2.4}$$

Lemma 3. *The lower bound in* (1.2) *holds.*

Proof. Assign the weights

$$\begin{gathered} w = \frac{1}{5}, \qquad u = v = \frac{1}{10}, \qquad a_i = b_i = \frac{1}{5n}, \\ d_j = \frac{1}{15n}, \qquad c = \frac{n-1}{5n}. \end{gathered} \tag{2.5}$$

Their sum is one. Moreover,

$$A = B = c + D = \frac{1}{5}, \qquad b_i + c = B - b_i + D = a_{i+1} + c = A - a_{i+1} + D = \frac{1}{5}.$$

Thus $P_1 = P_2 = 5^{-3}$. The first term in (2.4) is 5^{-5}, and the second is

$$\frac{1}{100(15n)^3} = \frac{5^{-5}}{4(3n)^3}.$$

□

3 A bound for open paths

The path estimate uses the following inequality for finite sequences.

Lemma 4. *Let $N \geq 1$, and let $x_0, \ldots, x_N$ be nonnegative real numbers such that $x_N = 0$ and $\sum_{i=0}^{N-1} x_i \geq 1/2$. Then*

$$\sum_{i=0}^{N-1} x_i(1 - x_{i+1})^2 \geq \frac{2}{9}. \tag{3.1}$$

Proof. Define

$$\psi(x) = \begin{cases} x - x^2/2, & 0 \leq x \leq 1, \\ 1/2, & x > 1. \end{cases}$$

For all $x, y \geq 0$,

$$\psi(x) - \psi(y) \leq x(1 - y)^2. \tag{3.2}$$

For $x, y \leq 1$, this follows from the identity

$$x(1-y)^2 - \psi(x) + \psi(y) = \frac{(x - 2y + y^2)^2 + y(1-y)^2(2-y)}{2} \geq 0.$$

If $y \geq 1$, the left side of (3.2) is nonpositive. If $x > 1 > y$, it equals $(1-y)^2/2$, so the inequality holds in this case as well.

If some $x_j \geq 1/3$, summing (3.2) from j to $N-1$ gives

$$\sum_{i=0}^{N-1} x_i(1 - x_{i+1})^2 \geq \psi(x_j) - \psi(0) \geq \frac{5}{18} > \frac{2}{9}.$$

Otherwise every $x_i < 1/3$, and the sum is at least $\frac{4}{9}\sum_{i=0}^{N-1} x_i \geq 2/9$. □

Lemma 5. *Let $L \geq 1$, and let $a_1, \ldots, a_L, b_1, \ldots, b_L, c, D$ be nonnegative numbers. Put*

$$A = \sum_{i=1}^{L} a_i, \qquad B = \sum_{i=1}^{L} b_i, \qquad a_{L+1} = 0,$$

and suppose that $A + B + c + D = 1$. Define the path forms

$$P_1 = \sum_{i=1}^{L} a_i(b_i + c)(B - b_i + D), \qquad P_2 = \sum_{i=1}^{L} b_i(a_{i+1} + c)(A - a_{i+1} + D).$$

Then, for every $s \in [0, 1]$,

$$sP_1 + (1-s)P_2 \leq \frac{1}{27} - \frac{s(1-s)D^3}{10}. \tag{3.3}$$

Proof. Set

$$\delta_A = A - \frac{1}{3}, \qquad \delta_B = B - \frac{1}{3}, \qquad S = |\delta_A| + |\delta_B|,$$

and

$$\alpha = \frac{A + D - c}{2} = D + \delta_A + \frac{\delta_B}{2}, \qquad \beta = \frac{B + D - c}{2} = D + \frac{\delta_A}{2} + \delta_B.$$

Completing the square gives

$$P_1 = \frac{A(1-A)^2}{4} - \sum_i a_i(b_i - \beta)^2, \tag{3.4}$$

$$P_2 = \frac{B(1-B)^2}{4} - \sum_i b_i(a_{i+1} - \alpha)^2. \tag{3.5}$$

Write

$$E = s\sum_i a_i(b_i-\beta)^2 + (1-s)\sum_i b_i(a_{i+1}-\alpha)^2, \qquad \Delta = \frac{1}{27} - sP_1 - (1-s)P_2.$$

The exact identity

$$\Delta = \frac{s\delta_A^2(4/3-A)}{4} + \frac{(1-s)\delta_B^2(4/3-B)}{4} + E$$

and the bounds $A, B \le 1-D$, $D \le 1$ imply

$$\Delta \ge \frac{D}{3}(s\delta_A^2 + (1-s)\delta_B^2) + E.$$

Indeed, $(4/3-A)/4 \ge (1/3+D)/4 \ge D/3$, and similarly for B. By the weighted Cauchy–Schwarz inequality,

$$\Delta \ge \frac{s(1-s)DS^2}{3} + E. \tag{3.6}$$

If $S \ge 11D/20$, then

$$\Delta \ge \frac{121}{1200}s(1-s)D^3 \ge \frac{s(1-s)D^3}{10},$$

as required. This includes $D = 0$, since $S \ge 0$. Suppose therefore that $S < 11D/20$, which implies $D > 0$. Since $D \le 1/3 + S$, we have

$$D \le \frac{20}{27}, \qquad A + B \ge \frac{2}{3} - S \ge \frac{7}{27} > \frac{1}{4}, \qquad 0 < D - S \le \alpha, \beta \le \frac{1}{2}.$$

Apply Lemma 4 to the sequence

$$\frac{a_1}{\alpha}, \frac{b_1}{\beta}, \ldots, \frac{a_L}{\alpha}, \frac{b_L}{\beta}, 0.$$

Its sum is $A/\alpha + B/\beta \ge 2(A+B) > 1/2$. Each successive pair x, y contributes to E a term $s\alpha\beta^2 x(1-y)^2$ or $(1-s)\beta\alpha^2 x(1-y)^2$. Consequently

$$E \ge \frac{2}{9}\min\{s\alpha\beta^2, (1-s)\beta\alpha^2\} \ge \frac{2}{9}s(1-s)(D-S)^3. \tag{3.7}$$

For all $S, m \ge 0$ we have

$$\frac{(S+m)S^2}{3} + \frac{2m^3}{9} \ge \frac{(S+m)^3}{10}. \tag{3.8}$$

Multiplying the difference by 90 gives the nonnegative expression

$$\begin{aligned} &21S^3 + 3S^2m - 27Sm^2 + 11m^3 \\ &= 21\left(S - \frac{3m}{5}\right)^2\left(S + \frac{6m}{5}\right) + 3m\left(S - \frac{18m}{25}\right)^2 + \frac{233m^3}{625}. \end{aligned}$$

Taking $m = D - S$ in (3.8) and using (3.6)–(3.7) proves (3.3). □

For $L \geq 1$, define the 5-uniform hypergraph T_L as in (2.1), with each index ranging over $1, \ldots, L$ and with the last link in H_2 replaced by the triples

$$\{B_L, C, Y\}, \qquad Y \in \{A_1, \ldots, A_L\} \cup \mathcal{D}.$$

The edge $\{U, V\} \cup \mathcal{D}$ is retained. Its polynomial is (2.4) with $a_{L+1} = 0$.

Lemma 6. *For every $L \geq 1$, we have $\lambda(T_L) = 5^{-5}$.*

Proof. The lower bound follows by assigning weight 1/5 to the vertices of any one edge and zero elsewhere. For the upper bound, take nonnegative vertex weights of total mass one, and put

$$h = w, \qquad q = u + v, \qquad t = A + B + c + D.$$

Thus $h + q + t = 1$. If $q = 0$ or $t = 0$, then $P_{T_L} = 0$. Otherwise put

$$s = u/q, \qquad d = D/t, \qquad z = s(1-s)d^3 \in [0, 1/4].$$

Applying Lemma 5 to the bottom weights divided by t, and then using homogeneity, gives

$$uP_1 + vP_2 \leq qt^3 \left(\frac{1}{27} - \frac{z}{10} \right).$$

The arithmetic–geometric mean inequality now yields

$$P_{T_L} \leq hqt^3 \left(\frac{1}{27} - \frac{z}{10} \right) + q^2 s(1-s) \left(\frac{td}{3} \right)^3. \tag{3.9}$$

Also,

$$\frac{qt^3}{27} \leq \left(\frac{q+t}{4} \right)^4, \qquad \frac{q^2 t^3}{3^3} \leq 4 \left(\frac{q+t}{5} \right)^5. \tag{3.10}$$

Define

$$F(h) = \frac{3125}{256} h(1-h)^4.$$

We have $F(h) \leq 1$, with equality at $h = 1/5$. Since $1 - 27z/10 > 0$, we may substitute (3.10) into (3.9). We get

$$\begin{aligned} \frac{P_{T_L}}{5^{-5}} &\leq F(h) \left(1 - \frac{27}{10} z \right) + 4s(1-s)d^3(1-h)^5 \\ &= F(h) + z \left[4(1-h)^5 - \frac{27}{10} F(h) \right]. \end{aligned} \tag{3.11}$$

Since $0 \leq 4z \leq 1$, the last expression in (3.11) is the convex combination

$$(1 - 4z)F(h) + 4z \left[\left(1 - \frac{27}{40} \right) F(h) + (1-h)^5 \right].$$

As $F(h) \leq 1$, it suffices to prove

$$\left(1 - \frac{27}{40} \right) F(h) + (1-h)^5 \leq 1 \qquad (0 \leq h \leq 1). \tag{3.12}$$

Since $(1 - 27/40)(3125/256) = 8125/2048 < 5$, the binomial expansion gives

$$\left(1 - \frac{27}{40} \right) F(h) + (1-h)^5 \leq 5h(1-h)^4 + (1-h)^5 \leq \big(h + (1-h) \big)^5 = 1.$$

This completes the upper bound. $\square$

4 The Lagrangian bound and the blow-ups

Proof of Proposition 2. The lower bound was proved in Lemma 3. For the upper bound, take any nonnegative weights on G_n of total mass one. Choose an index k with

$$a_k \le \frac{A}{n} \le \frac{1}{n}.$$

After deleting A_k, put $a'_1 = a'_{n+1} = 0$, $a'_j = a_{k+j-1}$ for $2 \le j \le n$, and $b'_j = b_{k+j-1}$ for $1 \le j \le n$, with indices modulo n. Writing $A' = \sum_j a'_j$, $B' = \sum_j b'_j$, the remaining polynomial is

$$\begin{aligned} P_{G_n - \{A_k\}} &= wu \sum_{j=1}^{n} a'_j (b'_j + c)(B' - b'_j + D) \\ &\quad + wv \sum_{j=1}^{n} b'_j (a'_{j+1} + c)(A' - a'_{j+1} + D) + uv \prod_{\ell=1}^{3} d_\ell \\ &= P_{T_n}\big|_{a'_1 = 0}. \end{aligned} \tag{4.1}$$

Thus Lemma 6 and homogeneity bound the total weight of the edges that remain by

$$5^{-5}(1 - a_k)^5 \le 5^{-5}.$$

For any simple 5-uniform hypergraph and weights of total mass one,

$$\frac{\partial P_H}{\partial x_v} \le \sum_{\substack{S \subseteq V(H)\setminus\{v\} \\ |S|=4}} \prod_{w \in S} x_w \le \frac{\left(\sum_{w \ne v} x_w\right)^4}{4!} \le \frac{1}{4!}. \tag{4.2}$$

Indeed, each product of four distinct weights occurs 4! times in the power expansion, and all other terms are nonnegative. The deleted edges consequently have total weight at most

$$a_k \frac{\partial P_{G_n}}{\partial a_k} \le \frac{a_k}{4!} \le \frac{1}{4!n}.$$

This proves (1.2). □

Proof of Theorem 1. Fix $\gamma > 5^{-5}$, and choose an integer $n \ge 2$ such that

$$5^{-5} + \frac{1}{4!n} < \gamma.$$

Keep n fixed, and set

$$\varepsilon = \frac{1}{4(3n)^3} > 0.$$

For each positive integer M, form a blow-up $B_{n,M}$ of G_n: replace each vertex by an independent class, and each edge by all transversal edges across its five classes. Use the following class sizes, with one class for each listed vertex:

vertex	W	U, V	A_i, B_i	C	D_j
class size	$6nM$	$3nM$	$6M$	$6(n-1)M$	$2M$

There are $N = 30nM$ vertices. These class sizes are proportional to the weights in (2.5), and the computation in Lemma 3 gives the exact edge count

$$e(B_{n,M}) = \left[(6n)^5 + 72n^2\right]M^5 = (1+\varepsilon)(N/5)^5. \tag{4.3}$$

Let S be any nonempty set of m vertices of $B_{n,M}$, and let k_v be the number of its vertices in the class corresponding to $v \in V(G_n)$. Then

$$e(B_{n,M}[S]) = \sum_{e\in E(G_n)} \prod_{v\in e} k_v = m^5 P_{G_n}\big((k_v/m)_v\big) \le m^5 \lambda(G_n) < \gamma m^5.$$

Together with (4.3) and $M \to \infty$, this proves the theorem. □

Corollary 7. *For every integer $r \ge 5$ and every $\gamma > r^{-r}$, there is an $\varepsilon > 0$ and there are arbitrarily large r-uniform hypergraphs H such that*

$$e(H) \ge (1+\varepsilon)\big(v(H)/r\big)^r, \qquad e(H[S]) < \gamma|S|^r \quad \textit{for every nonempty } S \subseteq V(H).$$

In particular, Problem 1075 fails for every $r \ge 5$.

Proof. We implement Peng's lifting construction [6, proof of Theorem 1, p. 763] by adjoining vertices before taking a blow-up. Let $q = r - 5$. For a 5-uniform hypergraph G, form an r-uniform hypergraph G^{+q} by adjoining the same set of q new vertices to every edge. If the total weight on $V(G)$ is t, then the arithmetic–geometric mean inequality gives

$$P_{G^{+q}} \le \lambda(G) t^5 \left(\frac{1-t}{q}\right)^q,$$

with the usual interpretation when $q = 0$. The right side is maximized at $t = 5/r$, and equality is attained by taking an optimal weighting of G, scaled by $5/r$, and weight $1/r$ on each new vertex. Hence

$$\lambda(G^{+q}) = \frac{5^5}{r^r}\lambda(G). \tag{4.4}$$

Applying (4.4) to G_n and using Proposition 2 gives

$$r^{-r}\left(1 + \frac{1}{4(3n)^3}\right) \le \lambda(G_n^{+q}) \le r^{-r} + \frac{5^5}{r^r 4! n}.$$

Choose n so that the upper bound is less than γ. For each positive integer M, multiply every class size in the proof of Theorem 1 by 5, and give each of the q new vertices a class of size $30nM$. These are integer class sizes for a blow-up H of G_n^{+q}, with

$$\begin{aligned} N &= 30nrM, \\ e(H) &= 5^5\left[(6n)^5 + 72n^2\right](30n)^{r-5}M^r = (1+\varepsilon)(N/r)^r, \end{aligned} \qquad \varepsilon = \frac{1}{4(3n)^3}.$$

The induced-subgraph argument in the proof of Theorem 1 gives $e(H[S]) \le \lambda(G_n^{+q})|S|^r < \gamma|S|^r$. Letting $M \to \infty$ proves the claim. □

Concluding remarks

The direct construction with $r-4$ common W-vertices and $r-2$ vertices in $\mathcal{D}$ has gain of order D^{r-2}; at $r=5$ this matches the cubic open-path loss, and Lemma 6 shows that the constants suffice to absorb it. The analogous power count at $r=4$ is heuristic: a quadratic gain would have to be absorbed by a cubic loss as $D \downarrow 0$. At $r=3$ the template does not exist, since a link edge $\{U\} \cup e$ already has four vertices. The endpoint question for $r=3,4$ remains open.

Together with Erdős's theorem that every number below $r!/r^r$ is a jump [2], Corollary 7 identifies $r!/r^r$ as the smallest non-jump for every $r \geq 5$. Mubayi raised in correspondence the question of whether this is the only transition for $r \geq 5$: are all numbers in $[r!/r^r, 1)$ non-jumps, or do jumps occur again above the first non-jump?

For $r=3$, Liu and Mubayi conjecture that every number in $[0, 4/9)$ is a jump and every number in $[4/9, 1)$ is a non-jump [9, Conjecture 1.2]. If this conjecture holds, the smallest non-jump for 3-graphs would be twice the Erdős threshold, whereas for every $r \geq 5$ it equals that threshold. Whether this dichotomy between $r=3$ and $r \geq 5$ is real, and what happens at $r=4$, is a natural next question.

Data and code availability

The Lean source, manuscript source and build instructions are publicly available at `https://github.com/FireflySentinel/erdos-1075`.

The formalization cited here is commit `109e271d0bad`, using Lean 4.33.0 and Mathlib 4.33.0. It covers the explicit hypergraphs and integer class sizes, the potential, cubic and scalar estimates used in Lemmas 4–6, the cycle bound in Proposition 2, and the blow-up and lifting arguments proving Theorem 1 and Corollary 7. The final theorem, `erdos1075_counterexamples`, is in namespace `Erdos1075` in `Erdos1075/Main.lean`. The repository README maps the proof steps to their Lean declarations.

The project contains no `sorry` or additional axioms. For the final theorem, `#print axioms` reports exactly

`[propext, Classical.choice, Quot.sound]`.

These are Lean's standard axioms of propositional extensionality, choice and quotient soundness. The cited revision passed continuous integration, including compilation, the axiom checks and kernel replay with `leanchecker`. No empirical datasets were generated or analyzed.

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Declaration of generative AI and AI-assisted technologies in the writing process

GPT-6 Astra proposed the cyclic construction and the Lagrangian estimates, and generated the manuscript and the Lean formalization. GPT-5.6 Sol and Claude Opus 5 were used for editorial review. The mathematical arguments, the Lean formalization and its correspondence with the manuscript were checked by the author, who takes responsibility for the content.

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